

Cover Letter

To: Editorial Office, *Quantum Science and Technology* (IOP Publishing)

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Dear Editors,

We are submitting “*Three Axes of Quantum Risk: A Unified Observability Model for PQC, QKD, and Crypto-Agility*” for consideration in *Quantum Science and Technology*.

PQ-readiness scanners today answer one question: is the primitive quantum-resistant? Two assets that look identical under that question can have very different migration costs, and very different channel-level protections, neither of which the scanner sees. The paper treats quantum-risk posture as a three-axis problem instead, scoring each asset on algorithmic resistance, channel protection, and migration agility, then folding the three into a deadline-adjusted risk score q . We validate all three axes with reproducible studies — a Tranco-top-1k TLS survey (43.8% hybrid PQ key exchange), a controlled ETSI GS QKD 014 emulator study (13 of 13 induced link/KME transitions classified correctly), and an 11-project crypto-agility pilot (91% agreement with hand-graded ground truth) — and release the reference implementation end to end: eBPF-based TLS observation, a working ETSI GS QKD 014 collector with a Key-Management-Entity emulator, and a static crypto-agility scanner. Every artifact of the ree0xQ platform (formerly *Sezar*) is open at <https://github.com/e2esolutions-tech/ree0xQ> under MIT.

Why QST specifically. The QKD axis treats channel-level quantum-secure key delivery as a measured property of an asset, not as a separate policy track, and the emulator lets practitioners reproduce the channel-state classification work without QKD hardware. We hope this combination — a measurable QKD-aware posture model paired with a reproducible test rig — is useful to QST readers working on the deployment side of the field.

Submission disclosures

- The manuscript is not under consideration at any other journal. A preprint is available on Research Square.
- Both authors approved the manuscript and the CRediT contributions stated in §10.
- We declare no competing interests. E2E Solutions develops and releases ree0xQ under MIT but derives no direct commercial revenue from the publication.
- The work was self-funded by E2E Solutions and Hacettepe University; no external grants.
- If accepted, we will reformat against the IOP iopart.cls template at the production stage.

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